

Insight Memo: Discount Policy Is Turning Profitable Orders Into Losses

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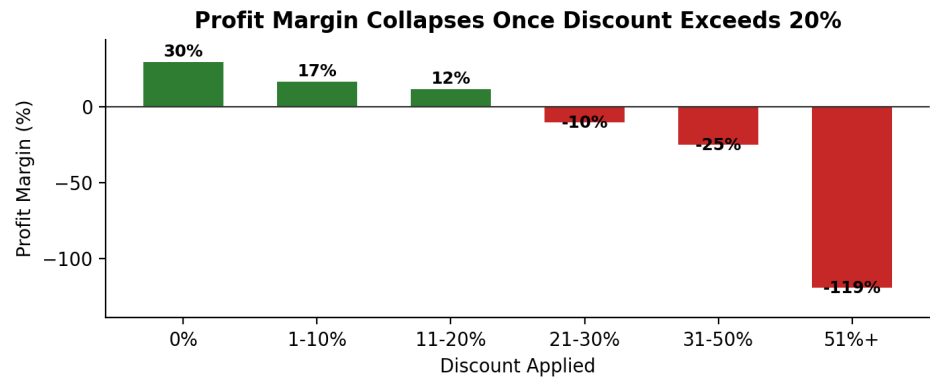
Context

Reviewed the Superstore transaction dataset (9,994 line items, 2014–2017) to assess whether discounting practices are protecting or eroding profitability across product lines.

Key Finding

Profit margin turns negative once a line item's discount exceeds 20%, and the collapse accelerates sharply from there — orders discounted above 50% lose more than the sale price itself:

- ▶ 0% discount: +29.5% margin | 11–20% discount: +11.6% margin
- ▶ **21–30% discount: –10.0% margin | 31–50%: –24.8% margin | 51%+: –119.2% margin**
- ▶ Orders discounted above 20% generated \$362,770 in sales but produced a combined –\$135,376 in losses (–37.3% margin on that slice of the business).



Sharpest Example: Tables

The Tables sub-category is the clearest single case of this pattern: \$206,966 in sales produced a net loss of \$17,725 (–8.6% margin), at an average discount of 26.1% — squarely inside the loss zone identified above. Bookcases (–3.0% margin) and Supplies (–2.5% margin) show the same pattern at smaller scale.

Business Impact

Roughly 1 in 5 line items (18.7%) currently operate at a loss, contributing –\$156,131 against the business's total profit of \$286,397. The discount-margin relationship above suggests this is a policy-driven, addressable problem rather than a demand or cost issue — every discount band under 20% remains solidly profitable, so the loss is concentrated and specific rather than systemic.

Recommendation

Cap discounting on Tables (and similarly loss-making sub-categories) below the 20% threshold where margin turns negative, or require a documented business justification for exceptions. Monitoring margin by discount band — not just by category or region — would catch this pattern earlier in future reporting cycles.

Methodology

Analysis performed in Python (Pandas) on the public Superstore sample dataset. Full code, dataset, and the original Power BI dashboard are available at github.com/Mustaqeem3313/Sales-Analysis-Dashboard.